**Math 550, S21** **Name:** Answer

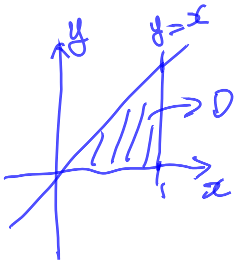
**Quiz 7**

__Instructions:__ **There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.**

**Problem 1.** (6 points) Evaluate the iterated integral $\int_0^1 \int_0^z \int_0^y 2ze^{-y^2}\, dxdydz$.

$$\int_0^1\int_0^z\int_0^y 2ze^{-y^2}dxdydz$$
$$=\int_0^1\int_0^z 2ze^{-y^2}[x]_{x=0}^{x=y}dydz = \int_0^1\int_0^z 2zye^{-y^2}dydz$$
$$=\int_0^1 z[-e^{-y^2}]_{y=0}^{y=z}dz = \int_0^1 z[1-e^{-z^2}]dz$$
$$=[\frac{z^2}{2}+\frac{e^{-z^2}}{2}]_{z=0}^{z=1} = \frac{1}{2}+\frac{1}{2}e^{-1}-0-\frac{1}{2}$$
$$=\frac{1}{2e}$$

**Problem 2.** (8 points) Evaluate the triple integral $\iiint_W 6xy\, dV$ where $W$ the region which lies under the plane $z = 1 + 2x + y$ and above the region in the $xy$-plane enclosed by the curves $y = x$, $y = 0$, and $x = 1$.

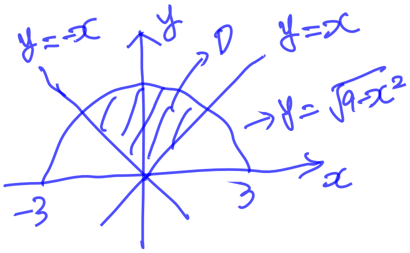



$$D=\{(x,y)\mid 0\le x\le 1,\ 0\le y\le x\}$$
$$W=\{(x,y,z)\mid (x,y)\in D,\ 0\le z\le 1+2x+y\}.$$

Thus

$$\iiint_W 6xy\,dV=\int_0^1\int_0^x\int_0^{1+2x+y}6xy\,dzdydx.$$
$$=\int_0^1\int_0^x 6xy[z]_{z=0}^{z=1+2x+y}dydx$$
$$=\int_0^1\int_0^x 6xy+12x^2y+6xy^2dydx.$$
$$=\int_0^1[3xy^2+6x^2y^2+2xy^3]_{y=0}^{y=x}dx$$
$$=\int_0^1 3x^3+8x^4dx$$
$$=[\frac{3}{4}x^4+\frac{8}{5}x^5]_{x=0}^{x=1}$$
$$=\frac{47}{20}$$

**Problem 3.** (8 points) A lamina covering the region $D$ in the plane is bounded by the curves $y = x$, $y = -x$ and $y = \sqrt{9 - x^2}$. Assume it has a density $f(x, y) = \sqrt{x^2 + y^2}$. Find the mass of the lamina by evaluating the double integral $\iint_D \sqrt{x^2 + y^2}\, dA$ in the **polar coordinates**.




$D^* = \{(r, \theta) \mid 0 \le r \le 3,\ \frac{\pi}{4} \le \theta \le \frac{3\pi}{4}\}.$

$$\iint_D \sqrt{x^2+y^2}\, dx\, dy$$

$$= \iint_{D^*} \sqrt{(r\cos\theta)^2 + (r\sin\theta)^2}\; r\, dr\, d\theta$$

$$= \int_0^3 \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} r^2\, d\theta\, dr$$

$$= \int_0^3 r^2 [\theta]_{\theta=\frac{\pi}{4}}^{\theta=\frac{3\pi}{4}}\, dr$$

$$= \frac{\pi}{2} \int_0^3 r^2\, dr$$

$$= \frac{\pi}{2} \left[\frac{r^3}{3}\right]_{r=0}^{r=3}$$

$$= \frac{9}{2}\pi$$